Optimal Control of Payload's Skew Rotation in Crane Systems with State and Control Input Constraints*

Ho Duc Tho, Ryosuke Tasaki, Takanori Miyoshi, and Kazuhiko Terashima

***Abstract*— In this paper, a nonlinear sub-optimal time control is proposed as a vibration controller for the payload's skew rotation process of crane systems. Physical constraints of the actuator including bounded velocity and bounded acceleration are explicitly taken into account in the design process. Clipping-off conjugate gradient method is adopted to solve the bounded optimal control problem, whereas penalty function is employed to relax the state constraint. Both simulation and experimental results are provided to clarify the effectiveness of the optimal control system. The proposed scheme can be directly applied to transfer the payload to desirable skew orientation without any residual oscillation, or it can be utilized as a reference trajectory planner of the skewing control module for a 2-DOFs controller in either overhead or rotary crane systems.**

I. INTRODUCTION

Skew orientation control of the payload plays an important role in crane systems. For instance, consider the overhead crane system as shown in Fig. 1 where the payload's center of gravity G has arrived at the correct *position* in the workspace without any oscillation. However, when the target *orientation* is taken into account, the transferring process is said to be incomplete since there exists a skewing mismatch angle β between the payload's orientation and its desirable pose. Therefore, it is essential to rotate the payload a skew angle of β along the Z axis in the counter-clockwise direction to accomplish the transferring task. A similar issue is also encountered in the boom crane systems in use at harbors where the skew orientation of the container must be adjusted to comply with that of the truck or vessel [1].

In practice, there are two types of mechanism currently being used to control the skew orientation of the payload, namely indirect and direct type. The indirect mechanism is usually utilized in the container crane system in which a sophisticated four-ropes configuration in conjunction with the usage of a spreader is presented [2]. Four actuators are equipped to independently control the length of each rope therefore it is able to control full orientations of the payload, namely trim-list-skew angles. However, in this type of mechanism, the maximum skewing angle that can be adjusted is only up to 10 degrees in both clockwise/counter-clockwise directions. Therefore, the flexibility of the skew adjustment is very limited. The second kind of the skewing control mechanism is illustrated in Fig. 1, where two-ropes configuration in combination with a simple device called

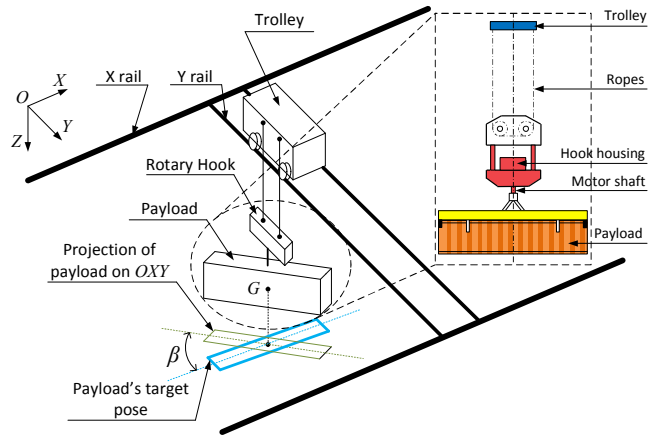

Fig. 1. Necessity of the payload orientation control in an overhead crane system. To adjust the payload's skew orientation, the direct mechanism with an aid of the rotary hook device is employed.

rotary hook are employed. In contrast to the indirect type, the rotary hook device is not designed to control the rope lengths. Alternatively, it will directly control the skewing orientation of the payload by a servo motor which is mounted inside the device. As shown in the zooming area in the Fig. 1, the payload is directly attached to the motor shaft and all of the suspended load are hung below the trolley through two flexible ropes. By using this kind of direct mechanism, the skewing orientation can be controlled to any desired value (> 360 degrees). However, along with the above advantage, after reaching the desired angle, the payload will continue to oscillate rather than stop at designated skew angle due to the law between inertia force and the rope tension. In practice, the payload inertia is large and the damping ratio of the system is low hence an artificial oscillation suppression needs to be imposed. At present, the payload orientation control through the rotary hook device is manually handled by highly experienced and skilled operator because of the under-actuated characteristic of the discussed system and as a result, the transfer process is not particularly time efficient. Therefore, an automatic controller should be developed to reduce the workload for the crane operator as well as drastically enhance the system productivity. Specifically, two control objectives need to be accomplished simultaneously, that is to say driving the payload to the desired skew angle and suppressing the residual skew oscillation caused by flexible ropes. Especially, this paper will focus on the direct mechanism type with the aid of rotary hook device to obtain a full flexibility in controlling the payload skew angle.

In the context where an overhead crane system is con-

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sidered, Sawodny et al. [3] introduce a linear model of a rotating system whose function is similar to the rotary hook device. Two degree of freedoms controller is designed based on the pole placement technique. Experimental results show that tracking error and vibration damping capacity of the proposed controller are satisfactory. However, a nonlinear model needs to be further developed and employed to enhance the control performance as well as the feasible operating region. Terashima et al. [4] establish a path planning and obstacle avoidance algorithm considering the rotary motion of the payload for an 3D overhead crane system. However, this study only emphasizes the potential method and its extension to solve the path generation problem. In the other words, suppression of the skew oscillation is not considered to an elaborate degree. In the circumstances regard to the boom crane system at the harbors, Gustafsson [5] formulates a nonlinear model of the rotary hook system in conjunction with the development of two controller types, namely time-optimal and state feedback control based on a simplified double integrator plant. Full-scaled experiments are conducted to verify the performance of the state feedback controller where the controller gains are chosen based on trial and error. Schaper et al. [6] propose a 2-DOFs skew control with a novel trajectory controller which consists of both state feedback and model predictive control feedforward paths. The performance of the control system is validated with test drives on a full-size mobile harbor boom crane. To improve the robust performance of the control system, an Adaptive Input Shaping (AIS) controller is formulated by the authors in [7]. The non-conventional algebraic estimator is the central of the AIS scheme which allows the Zero Vibration (ZV) input shaper robust to any variation range of natural frequency. Simulation studies are provided to prove the effectiveness of the AIS algorithm. In [1], a Sliding Mode Control (SMC) with a coupled linear sliding surface is proposed. Stability of the closed-loop system is proved based on the indirect Lyapunov method to obtain the algebraic inequality constraints of the SMC gains. The above study is developed in [8], [9] where a coupled integral sliding surface is employed as a replacement for the conventional linear type. Certain choices of the controller gains are adopted to guarantee the system slide on the sliding surface at the beginning of time as well as to be asymptotically stable in sense of the closed-loop dynamic. Effectiveness of SMCs in [1], [8], [9] are demonstrated on a miniature rotary hook system, in which the results are satisfactory without any gain tuning activities. However, in the aforementioned studies, a minimum time transferring process (subject to the physical constraints of the real rotary hook system) is not considered, hence the productivity is not optimum. Furthermore, the feedback controllers require a sophisticated and expensive sensor system. For this reason, a feedforward scheme based on the optimal control technique will be formulated to obtain a low-cost control system while maintaining a precise skew transferring and an effective residual oscillation suppression.

Various optimal control methods have been established to address the payload's *position* control in which the payload is considered as a point mass [10]–[12]. However, none of the existing research realizes an optimal controller for the payload's *orientation* adjustment. Therefore, in this paper, an optimal control scheme will be proposed for the rotary hook system to drive the payload to the desirable skewing orientation without any residual oscillation. Indoor application for the overhead crane system is adopted as the context in which the rotary hook system is utilized hence the external disturbances can be neglected. Physical constraints including bounded velocity and bounded control input will be elaborately taken into account. To solve the bounded optimal control problem, the clipping-off Conjugate Gradient (CG) method [13]–[15] in conjunction with the usage of penalty functions are employed. Sub-optimal time solution can be obtained by extending the fixed terminal time problem through a bisection algorithm. The experimental results are provided to prove effectiveness of the optimal control system.

The remainders of this paper are listed as follows. Section II formulates a nonlinear model of the rotary hook system by the Euler-Lagrange approach. A brief review of the clipping-off CG method is given in Section III. Formulation of the sub-optimal time control is presented in Section IV. Both simulation and experimental results are illustrated in Section V. Finally, some conclusions and future works are drawn in Section VI.

II. System modeling and validation

This section is devoted to formulating the mathematical modeling of the rotary hook system which consists of a rotary hook device, payload, ropes, and crane tip. The skew residual oscillation resulting from the rotation of the payload is depicted in Fig. 2, where $\varphi(t)$ denotes the relative angle between the payload and the hook, whereas $\theta(t)$ and $\gamma(t)$ represent the absolute angles between the hook and the payload versus x axis respectively. Table I describes the definitions of all system parameters.

In Fig. 2, P refers to the conjunction point of the load and the hook. Denote the length of OP line as a. The steel ropes are assumed to be inelastic thus following holonomic constraint must hold

$$a^2 = l^2 - 2R^2(1 - \cos\theta). \tag{1}$$

The first and second derivatives of a term are given by

$$\dot{a} = -\frac{R^2}{a}\dot{\theta}\sin\theta, \tag{2}$$

$$\ddot{a} = \frac{1}{a}\left(-\dot{a}^2 - R^2\ddot{\theta}\sin\theta - R^2\dot{\theta}^2\cos\theta\right). \tag{3}$$

The total kinetic energy of the system comprises of the rotation energies of the payload and the hook as well as the translation energy of the overall mass along the vertical direction. Therefore, it can be written by

$$T = \frac{1}{2}I_L\dot{\gamma}^2 + \frac{1}{2}I_H\dot{\theta}^2 + \frac{1}{2}m\dot{a}^2. \tag{4}$$

The potential energy of the system is given as

$$V = -mga. \tag{5}$$

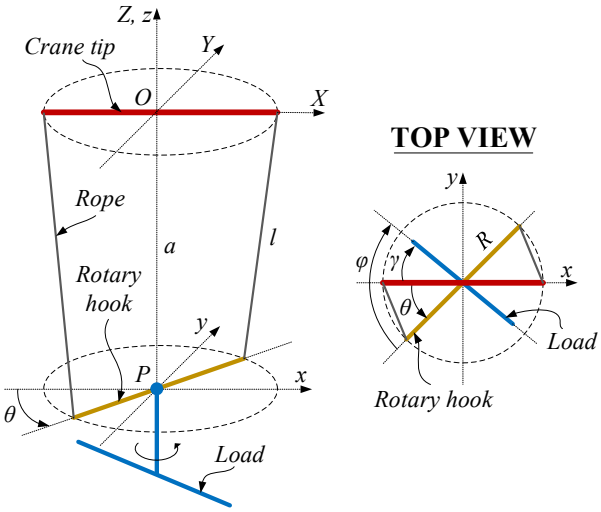

Fig. 2. Modeling of the rotary hook system.

TABLE I

PARAMETERS OF THE MINIATURE ROTARY HOOK SYSTEM.

Symbol	Description	Value
I_L	Inertia of the payload (kgm^2)	22.3
I_H	Inertia of the hook (kgm^2)	1.25
m	Total mass of the load and the hook (kg)	142.2
R	Skewing radius (m)	0.25
l	Rope length (m)	4
g	Gravity acceleration (m/s^2)	9.81

Therefore, the Lagrangian function can be established by

$$L = T - V = \frac{1}{2}I_L\dot{\gamma}^2 + \frac{1}{2}I_H\dot{\theta}^2 + \frac{1}{2}m\dot{a}^2 + mga. \quad (6)$$

The generalized coordinate vector of the system is chosen as $\boldsymbol{q} = \begin{bmatrix}\gamma(t) & \theta(t)\end{bmatrix}^\top$. By applying the Euler-Lagrange equation on the first generalized coordinate $\theta(t)$, we have

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{\theta}}\right) - \frac{\partial L}{\partial \theta} = 0. \quad (7)$$

Note that $\gamma(t) = \varphi(t) - \theta(t)$. Necessary results for developing (7) are given in (8)-(9)

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{\theta}}\right) = -I_L\ddot{\gamma} + I_H\ddot{\theta} + m\ddot{a}\frac{\partial \dot{a}}{\partial \dot{\theta}} + m\dot{a}\frac{d}{dt}\left(\frac{\partial \dot{a}}{\partial \dot{\theta}}\right), \quad (8)$$

$$\frac{\partial L}{\partial \theta} = m\dot{a}\frac{\partial \dot{a}}{\partial \theta} + mg\frac{\partial a}{\partial \theta}, \quad (9)$$

where

$$\frac{\partial \dot{a}}{\partial \dot{\theta}} = -\frac{R^2}{a}\sin\theta, \quad (10)$$

$$\frac{d}{dt}\left(\frac{\partial \dot{a}}{\partial \dot{\theta}}\right) = \frac{\dot{a}}{a^2}R^2\sin\theta - \frac{R^2}{a}\dot{\theta}\cos\theta, \quad (11)$$

$$\frac{\partial \dot{a}}{\partial \theta} = -\frac{1}{a^2}\frac{\partial a}{\partial \theta}(-R^2\dot{\theta}\sin\theta) - \frac{R^2}{a}\dot{\theta}\cos\theta, \quad (12)$$

$$\frac{\partial a}{\partial \theta} = -\frac{R^2}{a}\sin\theta, \quad (13)$$

$$\frac{\partial a}{\partial \dot{\theta}} = 0. \quad (14)$$

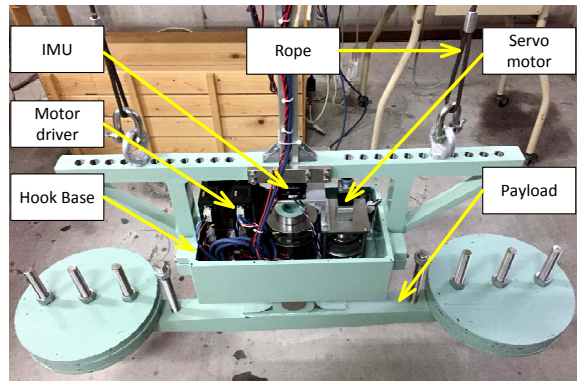

Fig. 3. Structure of the experimental testbed. A high accuracy 9-DOFs IMU is utilized to measure the skew oscillation angle $\theta(t)$.

Substituting (8)-(14) and (2)-(3) into (7) yields

$$\left(I_H + \frac{mR^4\sin^2\theta}{a^2}\right)\ddot{\theta} = -\frac{mR^2\sin\theta}{a}\left(\frac{R^4\dot{\theta}^2\sin^2\theta}{a^3} + \frac{R^2}{a}\dot{\theta}^2\cos\theta + g\right) + I_L\ddot{\gamma}. \quad (15)$$

Eq. (15) can be rearranged in a standard form as follows

$$\ddot{\theta} = f + bu, \quad (16)$$

where

$$f = \frac{1}{I_H + \frac{mR^4\sin^2\theta}{a^2}} \times \left[\frac{-mR^2\sin\theta}{a}\left(\frac{R^4\dot{\theta}^2\sin^2\theta}{a^3} + \frac{R^2\dot{\theta}^2\cos\theta}{a} + g\right)\right]$$

$$b = \frac{I_L}{I_H + \frac{mR^4\sin^2\theta}{a^2}}$$

$$u = \ddot{\gamma}, \quad (17)$$

and $a = \left(l^2 - 2R^2\left(1 - \cos\theta\right)\right)^{1/2}$.

It is remarked that in (16), control input of the system is a acceleration $\ddot{\gamma}(t)$ generated from the servo motor. In addition to that, there are two control objectives need to be achieved, namely $\gamma(t) \to \gamma_d$ and $\theta(t) \to 0$, where γ_d is the reference skew angle of the payload, thus clearly, rotary hook is an under-actuated system.

In order to verify the effectiveness of the proposed control schemes, an experimental apparatus is built, whose configuration is illustrated in Fig. 3. The actuator is a TAMAGAWA TS4738 AC servo motor. The incremental encoder attached at the end of the motor has a resolution of 2048×400 pules/rev. According to the motor and gear transmission ratio specifications, the maximum speed and acceleration capacities of the servo motor actuating on the payload are given as $v_{\max} = 0.524$ rad/s and $u_{\max} = 0.4$ (rad/s^2). Note that, the experimental apparatus is designed based on the real-size system in use at the harbor; therefore, it does not generate any difficulty in the mechanical property when the

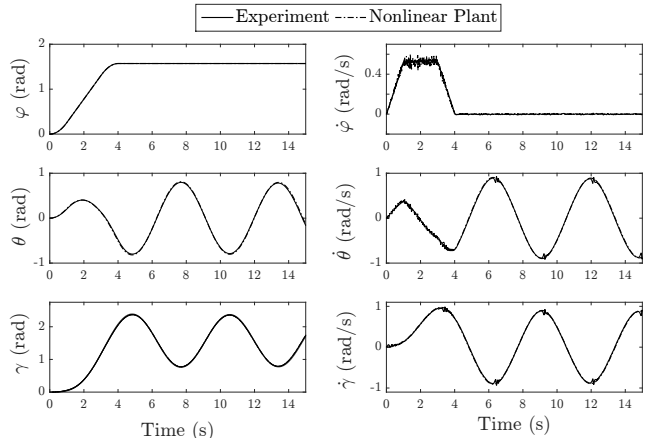

Fig. 4. Result of the model validation. The experimental data is in a total agreement with the mathematical model (16).

subsequent proposed controller is implemented on the full-size device.

The nonlinear model of the rotary hook system (16) now will be validated in the experiment to verify its correctness. The actual system parameters of the experimental testbed shown in Table I will be supplied for the nonlinear model given by (16) in the simulation. The result is shown in Fig. 4. It is observed that strong agreements between the experimental data and the mathematical model have been confirmed. Therefore it is concluded that the nonlinear model (16) accurately describe the dynamic behavior of the rotary hook system.

At this point, the mathematical modeling of the rotary hook system is successfully established. The optimal control design will be discussed in Section IV. In practice, the physical constraints of the actuator must be taken into account, thus in this paper, both control input and state constraints will be considered in the optimal control formulation. Generally, it is almost impossible to obtain an analytical solution of the constrained optimal control problem for a nonlinear system except some simple cases [16]–[18]. Therefore, numerical method will be adopted in this paper.

III. Preliminary of the Clipping-off Conjugate Gradient Method

In the domain of numerical approaches to solve a constrained optimal control problem, direct [11], [19], [20] and indirect [13]–[15] methods can be found in the literature. However, the indirect techniques always result in a faster convergence rate and better cost performance index [18], thus it will be employed in this paper. Quintana and Davison [15] introduced a novel conjugate gradient method in conjunction with a clipping-off operator to effectively solve the bounded optimal control problem. This technique elegantly incorporates the classical conjugate gradient method (for solving an unconstrained optimization problem) and the clipping-off operator to enforce the local optimizer satisfying the necessary condition of optimality subjected to the control input constraints. Therefore, the scheme naturally suits for resolving the optimal control problem posed in this paper [see (32)]. For the sake of convenience in following presentations, the algorithm of [15] will be briefly reviewed.

Consider the following nonlinear system

$$\dot{\boldsymbol{x}}(t) = \boldsymbol{f}(\boldsymbol{x}, \boldsymbol{u}) \quad \boldsymbol{x}(t_0) = \boldsymbol{x_0}, \tag{18}$$

where $\boldsymbol{f}$ is a nonlinear n-dimensional vector function of n-dimensional state vector $\boldsymbol{x}$ and r-dimensional control input $\boldsymbol{u} = [u_1 \;\; u_2 \;\; \ldots \;\; u_r]^\top$ (i.e. $\boldsymbol{f} \in \mathbb{R}^n$, $\boldsymbol{x} \in \mathbb{R}^n$, $\boldsymbol{u} \in \mathbb{R}^r$). Furthermore, t_0 denotes the initial time whereas t_f is the terminal time, which are assumed to be known. The optimal control problem with a fixed terminal time t_f is posed as

$$\begin{aligned} \min_{\boldsymbol{u}} \quad & J(\boldsymbol{u}) = \phi(\boldsymbol{x}(t_f), t_f) + \int_{t_0}^{t_f} L(\boldsymbol{x}, \boldsymbol{u}) dt \\ \text{s.t.} \quad & \dot{\boldsymbol{x}} = \boldsymbol{f}(\boldsymbol{x}, \boldsymbol{u}) \\ & \boldsymbol{u} \in \Omega, \end{aligned} \tag{19}$$

therein the admissible control set Ω is defined as

$$\Omega \triangleq \{u_j : |u_j| \leq u_{\max} \quad (j = 1, 2, \ldots, r)\}. \tag{20}$$

Denote H as the Hamiltonian function

$$H(\boldsymbol{x}, \boldsymbol{\lambda}, \boldsymbol{u}) \triangleq \boldsymbol{\lambda}^\top \boldsymbol{f}(\boldsymbol{x}, \boldsymbol{u}) + L(\boldsymbol{x}, \boldsymbol{u}), \tag{21}$$

where the n-dimensional costate vector $\boldsymbol{\lambda}(t)$ must satisfy

$$\dot{\boldsymbol{\lambda}}(t) = -\left(\frac{\partial H}{\partial \boldsymbol{x}}\right)^\top \text{ and } \boldsymbol{\lambda}(t_f) = \left(\frac{\partial \phi(\boldsymbol{x}(t_f), t_f)}{\partial \boldsymbol{x}(t_f)}\right)^\top. \tag{22}$$

By letting $\boldsymbol{g}(t) = \left(\frac{\partial H}{\partial \boldsymbol{u}}\right)^\top = \begin{bmatrix} \frac{\partial H}{\partial u_1} & \cdots & \frac{\partial H}{\partial u_r} \end{bmatrix}^\top \triangleq [g_1(t) \;\; \ldots \;\; g_r(t)]^\top$, following control input constraints must be fulfilled in order to guarantee the necessary condition for optimal solution [21]

$$\begin{cases} u_j = u_{\max} & \text{if } \dfrac{\partial H}{\partial u_j} < 0 \\ -u_{\max} \leq u_j \leq u_{\max} & \text{if } \dfrac{\partial H}{\partial u_j} = 0 \\ u_j = -u_{\max} & \text{if } \dfrac{\partial H}{\partial u_j} > 0 \end{cases}, \quad j = 1, \ldots, r. \tag{23}$$

Denote $\{\mathbb{T} = t : t \in [t_0 \;\; t_f]\}$. From (23), it is motivated to define two sub-interval points sets, namely "unsaturated-control sub-interval points" $\mathbb{U}_j^i$ and "saturated-control sub-interval points" $\mathbb{S}_j^i$ at ith iteration of jth control input as follows

$$\mathbb{S}_j^i = \left\{t : \left|u_j^i(t)\right| = u_{\max} \text{ and } \operatorname{sgn}(u_j^i(t)) = -\operatorname{sgn}(g_j^i(t))\right\}. \tag{24}$$

$$\mathbb{U}_j^i = \mathbb{T} - \mathbb{S}_j^i, \; t \in \mathbb{T}, \; j = 1, 2, \ldots, r, \tag{25}$$

where "sgn" represents an usual signum function.

Algorithm 1: Clipping-off conjugate gradient algorithm

Step 1 At every time step t, giving an initial estimate of bounded optimal control input $\boldsymbol{u}^0(t) = \left[u_1^0(t) \;\; \ldots \;\; u_r^0(t)\right]^\top \in \Omega$, compute $\boldsymbol{g}^0(t) =$

$\left(\frac{\partial H}{\partial \boldsymbol{u}^0}\right)^\top \triangleq \left[g_1^0(t) \ \ldots \ g_r^0(t)\right]^\top$ and let the initial search direction as $\boldsymbol{p}^0(t) = -\boldsymbol{g}^0(t)$.

Step 2 Choose a scalar step-size α^0 such that $J\left(\left\{\boldsymbol{u}^0 + \boldsymbol{\varepsilon}^0\boldsymbol{p}^0\right\}_{\boldsymbol{s}}\right)$ is minimized by means of any one dimensional linesearch method, where value of $\boldsymbol{\varepsilon}^0$ at the time step t is given as $\boldsymbol{\varepsilon}^0(t) = \left[\varepsilon_1^0(t) \ \ldots \ \varepsilon_r^0(t)\right]^\top$ and

$$\varepsilon_j^0(t) = \begin{cases} 0 & \text{if } t \in \mathbb{S}_j^0 \\ \alpha^0 > 0 & \text{if } t \in \mathbb{U}_j^0 \end{cases}, \ (j = 1, \ldots, r). \tag{26}$$

For clarification, $\left\{\boldsymbol{u}^0 + \boldsymbol{\varepsilon}^0\boldsymbol{p}^0\right\}_{\boldsymbol{s}}$ at t is given by

$$\left\{\boldsymbol{u}^0 + \boldsymbol{\varepsilon}^0\boldsymbol{p}^0\right\}_{\boldsymbol{s}} = \begin{bmatrix} \left\{u_1^0(t) + \varepsilon_1^0(t)p_1^0(t)\right\}_s \\ \vdots \\ \left\{u_r^0(t) + \varepsilon_r^0(t)p_r^0(t)\right\}_s \end{bmatrix}, \tag{27}$$

and

$$\{(.)\}_s = \begin{cases} (.) & \text{if } |(.)| \le u_{\max} \\ u_{\max} & \text{if } (.) > u_{\max} \\ -u_{\max} & \text{if } (.) < -u_{\max}. \end{cases} \tag{28}$$

Let $\boldsymbol{u}^1 = \left\{\boldsymbol{u}^0 + \boldsymbol{\varepsilon}^0\boldsymbol{p}^0\right\}_{\boldsymbol{s}}$ and $i = 1$.

Step 3 At each t, determine the next search direction $\boldsymbol{p}^i(t) = \left[p_1^i(t) \ \ldots \ p_r^i(t)\right]^\top$ as follows

$$p_j^i(t) = \begin{cases} -g_j^i(t) + \beta_j^i p_j^{i-1}(t) & \text{if } t \in \mathbb{U}_j^i \\ -g_j^i(t) & \text{if } t \in \mathbb{S}_j^i, \end{cases} \tag{29}$$

where $j = 1, \ldots, r$ and

$$\beta_j^i = \begin{cases} (\beta_0^i)_j & \text{if } (\beta_0^i)_j < (\beta_m^i)_j \\ \gamma^i(\beta_m^i)_j & \text{if } (\beta_0^i)_j \ge (\beta_m^i)_j, \ 0 < \gamma^i < 1 \\ 0 & \text{if } (I_2^i)_j = 0 \text{ or } (I_3^i)_j = 0, \end{cases} \tag{30}$$

with $(\beta_0^i)_j \triangleq \frac{(I_1^i)_j}{(I_2^i)_j}$, $(I_2^i)_j \neq 0$; $(\beta_m^i)_j \triangleq \frac{(I_1^i)_j}{(I_3^i)_j}$, $(I_3^i)_j \neq 0$, and

$$\begin{aligned} (I_1^i)_j &= \int_{t \in \mathbb{U}_j^i} g_j^i(t) g_j^i(t) dt \\ (I_2^i)_j &= \int_{t \in \mathbb{U}_j^i} g_j^{i-1}(t) g_j^{i-1}(t) dt \\ (I_3^i)_j &= \int_{t \in \mathbb{U}_j^i} g_j^i(t) p_j^{i-1}(t) dt. \end{aligned} \tag{31}$$

Step 4 Choose a step-size α^i to minimize $J\left(\left\{\boldsymbol{u}^i + \boldsymbol{\varepsilon}^i\boldsymbol{p}^i\right\}_{\boldsymbol{s}}\right)$ in the similar manner with step 2, then let $\boldsymbol{u}^{i+1} = \left\{\boldsymbol{u}^i + \boldsymbol{\varepsilon}^i\boldsymbol{p}^i\right\}_{\boldsymbol{s}}$ and $i \leftarrow i + 1$. Go to step 5.

Step 5 Repeat step 3 and step 4 until $\left|J(\boldsymbol{u}^{i+1}) - J(\boldsymbol{u}^i)\right| < \delta\left|J(\boldsymbol{u}^{i+1})\right|$ or $J(\boldsymbol{u}^{i+1}) < J_{tol}$ where $\delta > 0$ regards to tolerance at which the algorithm is said to be converged. Here, tolerance δ is chosen as $\delta = 10^{-6}$.

Remark 1: In the step 2 and 4, the minimizer α^i of $J\left(\left\{\boldsymbol{u}^i + \boldsymbol{\varepsilon}^i\boldsymbol{p}^i\right\}_{\boldsymbol{s}}\right)$ can be obtained by any one-dimensional linesearch method. In this paper, golden section method [22] will be employed. Specifically, a high-low-high geometry pattern of the cost functional J with respect to α^i should be found beforehand to determine the feasible searching interval. Furthermore, the J_{tol} presents in Step 5 can be chosen heuristically based on the requirements of the system's control performance.

By means of Algorithm 1, local minimizer of the optimal control problem (19) can be *always* obtained. Furthermore, all of necessary conditions for optimality presented in (22)-(23) are fulfilled by the resultant local optimum solution [15]. Note that, at all times, Algorithm 1 would converge to a local optimal solution. However, the quality of the converged solution depends on the terminal time t_f.

IV. Optimal control formulation and algorithm

In this section, optimal control formulation for the rotary hook system is presented where two control objectives, namely driving the payload to the desired skew angle and suppressing the skew oscillation simultaneously (i.e. forcing $\gamma(t) \to \gamma_d$ and $\theta(t) \to 0$) are the basics to establish the cost functional. Denote a state vector of the control system as $\boldsymbol{x} = \begin{bmatrix} x_1 & x_2 & x_3 & x_4 \end{bmatrix}^\top = \begin{bmatrix} \gamma(t) & \dot{\gamma}(t) & \theta(t) & \dot{\theta}(t) \end{bmatrix}^\top$. By taking the physical limitations of the actuator including the maximum velocity $v_{\max}$ and maximum acceleration $u_{\max}$ into account, the optimal control problem for the rotary hook system can be posed as

$$\begin{aligned} \min_u \ & J = \omega_1\left(x_1(t_f) - \gamma_d\right)^2 + \omega_2 x_2^2(t_f) + \omega_3 x_3^2(t_f) + \omega_4 x_4^2(t_f) \\ s.t. \ & \dot{\boldsymbol{x}} = \boldsymbol{f}(\boldsymbol{x}) + \boldsymbol{b}(\boldsymbol{x})u, \\ & (x_2 + x_4)^2 \le v_{\max}^2, \\ & |u| \le u_{\max}. \end{aligned} \tag{32}$$

In (32), t_f is the fixed terminal time and

$$\boldsymbol{f}(\boldsymbol{x}) = \begin{bmatrix} x_2 & 0 & x_4 & f \end{bmatrix}^\top \text{ and } \boldsymbol{b}(\boldsymbol{x}) = \begin{bmatrix} 0 & 1 & 0 & b \end{bmatrix}^\top, \tag{33}$$

where f and b terms are given in (17). The weighting coefficients are chosen as $\omega_1 = \ldots = \omega_4 = 10^6$. As discussion in Section II, $u_{\max} = |\ddot{\gamma}|_{\max} = 0.4$ (rad/s^2) and $v_{\max} = |\dot{\varphi}|_{\max} = |x_2 + x_4|_{\max} = 0.524$ (rad/s).

Note that the optimal control problem (32) is involved by both control input and state constraints. As discussed in Section III, control input constraint can be efficiently solved by the clipping-off CG method. To eliminate the state constraint, the original cost functional J will be augmented by a step penalty function. The state-constraint-free optimal control problem can be further described as

$$\begin{aligned} \min_u \quad & J_s = J + \underbrace{\int_{t_0}^{t_f} k_1 \phi_1^2 dt}_{\text{Penalty term}} \\ s.t. \quad & \dot{\boldsymbol{x}} = \boldsymbol{f}(\boldsymbol{x}) + \boldsymbol{b}(\boldsymbol{x})u \\ & |u| \le u_{\max} \\ & \phi_1 = v_{\max}^2 - (x_2 + x_4)^2 \\ & k_1 = \begin{cases} 0 & \text{if } \phi_1 \ge 0 \\ \mu_1 & \text{if } \phi_1 < 0, \end{cases} \end{aligned} \tag{34}$$

where $\mu_1 = 10^6$. Note that the penalty coefficient k_1 is activated whenever the state constraint represented in ϕ_1 is violated.

Denote $u^\star(t_f)$ as the resultant optimal control input of (34) with respect to the fixed terminal time t_f. The sub-optimal time control problem can be solved by extending (34) by means of the bisection method [12], which is described as follows.

Algorithm 2: Bisection method to solve the sub-optimal time problem.

Step 1 Let $t_f = \dfrac{\gamma_d}{\alpha v_{\max}}$, $a_1 \leftarrow 0$, and $a_2 \leftarrow t_f$. Run *Algorithm 1* to obtain the optimal control input $u^\star(t_f)$. If $J_s(u^\star, t_f) < J_{tol}$ then to step 2, otherwise go to step 3.

Step 2 Let $t_f \leftarrow (a_1 + a_2)/2$. Run Algorithm 1 to obtain $u^\star(t_f)$. If $J(u^\star, t_f) < J_{tol}$ then $a_2 \leftarrow t_f$, otherwise $a_1 \leftarrow t_f$, go to Step 4.

Step 3 Let $a_1 \leftarrow a_2$, $a_2 \leftarrow 2a_2$, and $t_f \leftarrow a_2$. Obtain $u^\star(t_f)$ from Algorithm 1. If $J_s(u^\star, t_f) > J_{tol}$ then repeat Step 3, otherwise go to Step 2.

Step 4 If $a_2 - a_1 > \Delta_t$ then go to Step 2. Otherwise, go to Step 5.

Step 5 If $J_s(t_f = a_1) < J_{tol}$ then $t_f \leftarrow a_1$, otherwise $t_f \leftarrow a_2$. Output $u^\star(t_f)$ as the optimal control input corresponding to the minimum transferring time t_f.

In Algorithm 2, α relates to the initial value of the terminal time t_f. Typically, it can be chosen that $\alpha = 1$. Moreover, J_{tol} specifies the requirements corresponding to the transfer accuracy, residual vibration suppression, and state constraint assurance. With the large weighting coefficients ($\omega_1 = \ldots = \omega_4 = 10^6$), it can be heuristically chosen that $J_{tol} = 1$ to sufficiently fulfill the control performance. Terminated condition of Algorithm 2 is determined by the tolerance Δ_t which regards to the minimum distance between two consecutive trial terminal time points. Here, $\Delta_t = 0.1$ second is adopted. It is important to remark that the initial guess of the optimal control input for the present terminal time t_f should be taken from that of the previous successful iteration. Such choice can significantly reduce the computational effort.

V. Simulation and Experimental Results

By employing Algorithm 2, optimal controls with both input and state constraints will be computed for three cases: short ($\gamma_d = 30$ degrees), medium ($\gamma_d = 90$ degrees), and long transferring case ($\gamma_d = 180$ degrees). Initial guess of the optimal control input in the clipping-off CG algorithm is given as $u(t) = 0, t \in \mathbb{T}$. The computation is conducted on a personal computer with 2.5-GHz Intel Core i5-3210M and 8 GB of RAM. The computational results for all cases are summarized in Table II. Note that the clipping-off CG method is restarted in every 2 iterations.

The simulation and experimental results in three cases are shown in Fig. 5 - Fig. 7 respectively. It can be recognized that with the computed optimal control inputs, the payload accurately reach the desired skew angle as well as the skew oscillation is nearly vanished after the motion. Both control input and state constraints are assured by the resultant optimal solution. Therefore it is concluded that the effectiveness of the proposed optimal control system is verified.

TABLE II

Computation results of the no-switched optimal controls.

Case	$\min(t_f)$	J_s	Compt. time	$\max(-\phi_1(t))$
$\gamma_d = 30°$	3.13 s	0.6651	35.1 s	0.0170
$\gamma_d = 90°$	5.06 s	0.8060	61.7 s	0.0093
$\gamma_d = 180°$	7.97 s	0.4165	85.2 s	0.0072

Compt. time: Computational time.

In comparison with the feedback control system which is established in [8], the proposed sub-optimal time controller performs much faster transferring time. For instance, when $\gamma_d = 90°$, it takes 5.06 seconds for the proposed feed-forward scheme to complete the transferring process without any residual vibration whereas 10.1 seconds is required by the feedback controller presented in [8]. Therefore, in case

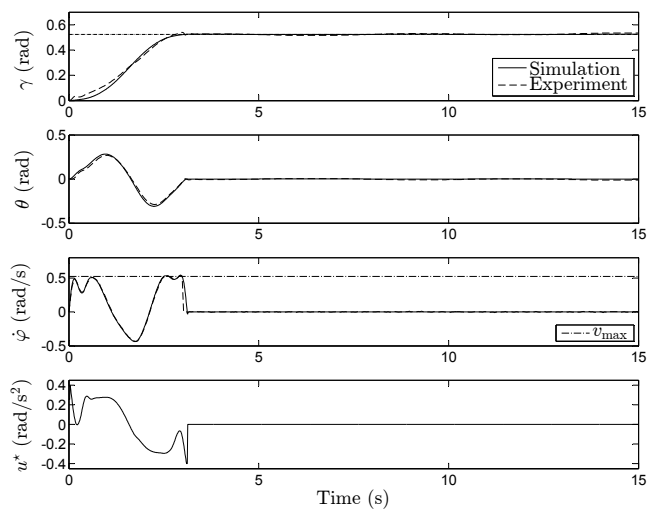

Fig. 5. Simulation and experimental results in the short transfer case.

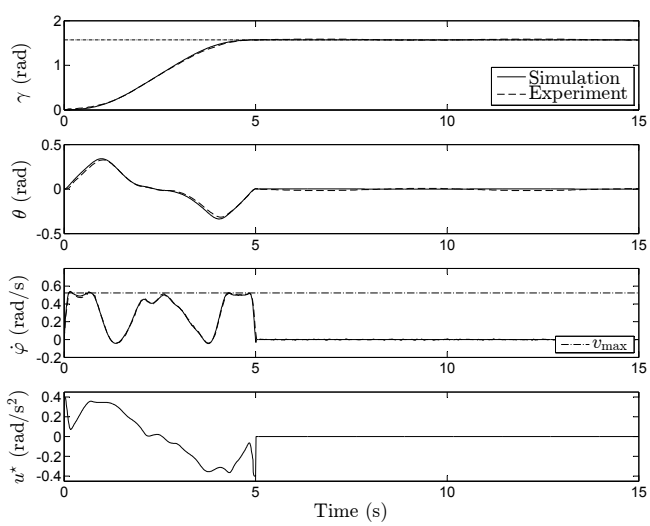

Fig. 6. Simulation and experimental results in the medium transfer case.

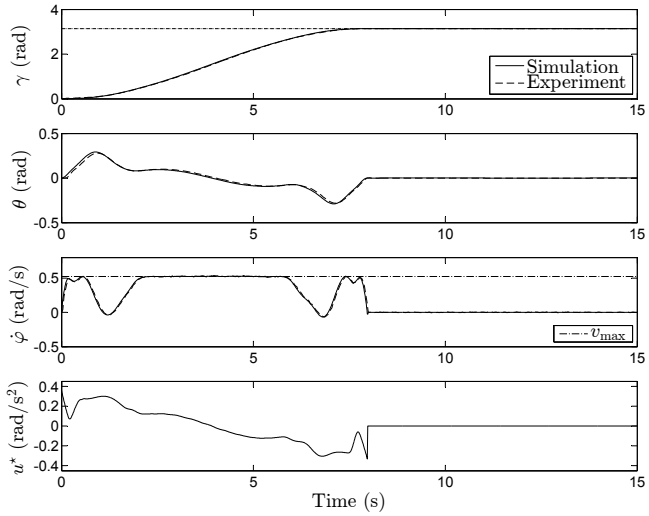

Fig. 7. Simulation and experimental results in the long transfer case.

that the system parameters do not significantly vary, the proposed scheme can be effectively utilized as a standalone controller. Otherwise, it could be used as an trajectory planner for the feedback control system to enhance the overall control performance.

Remark 2: In conjunction to the excellent property in resolving both state and control input constraints, the main shortcoming of any nonlinear optimal control method is the high computational burden, which turns out a significant challenge in the real-time computation. This issue is indicated in Table II, where the computational time of all cases are considerably high. However, to tackle such problem, one could off-line build a map (or lookup table) of the control input profiles for each target skew angle γ_d. Therefore, the upload-and-run strategy can be easily realized.

VI. Conclusions and Future Works

In this paper, a nonlinear model of the rotary hook system is formulated. The proposed mathematical modeling is then validated in the experiment. A sub-optimal time controller is established to drive the payload to the desirable skewing orientation in a minimum time without any residual oscillation. Both velocity and acceleration constraints, as a consequence of the servo motor's physical limitations, are explicitly considered in the design process. The experimental results are provided to prove the effectiveness of the optimal control system. Our future work is to utilize the functionality of an electro-mechanical clutch to reduce the energy consumption of the overall control system. In such scenario, a switched optimal system must be realized due to the engaging/disengaging operations of the clutch. Furthermore, the skew rotation dynamic of the rotary hook system will be integrated into the boom crane dynamic in order to design a comprehensive automatic controller which can suppress not only the swing angles but also residual skew oscillation of the payload simultaneously during the transfer process.

References

[1] H. D. Tho, R. Tasaki, K. Terashima, M. Tsume, and K. Suzuki, "Robust sliding mode control of a rotary hook," *IFAC-PapersOnLine*, vol. 50, no. 1, pp. 6060 – 6066, 2017, 20th IFAC World Congress.

[2] A. Arena, A. Casalotti, W. Lacarbonara, and M. P. Cartmell, "Dynamics of container cranes: three-dimensional modeling, full-scale experiments, and identification," *Int. J. of Mech. Sciences*, vol. 93, pp. 8 – 21, 2015.

[3] O. Sawodny, H. Aschemann, and S. Lahres, "An automated gantry crane as a large workspace robot," *Control Eng. Practice*, vol. 10, no. 12, pp. 1323 – 1338, 2002.

[4] K. Terashima and M. Suzuki, "Path planning and navigation of overhead traveling crane with three-dimensional transport based on a diffusion equation strategy," in *Advances in Manufacturing*. London: Springer London, 1999, pp. 335–347.

[5] T. Gustafsson, "Modelling and control of rotary crane syst." Ph.D. dissertation, Div. of Automat. Control, Lulea Univ. of Tech., Lulea, Sweden, 1993.

[6] U. Schaper, C. Dittrich, E. Arnold, K. Schneider, and O. Sawodny, "2-dof skew control of boom cranes including state estimation and reference trajectory generation," *Control Eng. Practice*, vol. 33, pp. 63 – 75, 2014.

[7] H. D. Tho, K. Takemori, R. Tasaki, T. Miyoshi, and K. Terashima, "Adaptive input shaping control of a rotary hook," in *2016 55th Annu. Conf. of the Soc. of Instrument and Control Engineers of Japan (SICE)*, Sept. 2016, pp. 596–602.

[8] H. D. Tho, R. Tasaki, A. Kaneshige, T. Miyoshi, and K. Terashima, "Robust sliding mode control with integral sliding surface of an underactuated rotary hook system," in *2017 IEEE Int. Conf. on Advanced Intelligent Mechatronics (AIM)*, July 2017, pp. 998–1003.

[9] H. D. Tho and K. Terashima, "Robust control designs of payload's skew rotation in a boom crane system," *IEEE Transactions on Control Systems Technology*, pp. 1–14, 2018.

[10] Y. Sakawa and Y. Shindo, "Optimal control of container cranes," *Automatica*, vol. 18, no. 3, pp. 257 – 266, 1982.

[11] A. Al-Garni, K. Moustafa, and S. J. Nizami, "Optimal control of overhead cranes," *Control Eng. Practice*, vol. 3, no. 9, pp. 1277 – 1284, 1995.

[12] K. Terashima, Y. Shen, and K. Yano, "Modeling and optimal control of a rotary crane using the straight transfer transformation method," *Control Eng. Practice*, vol. 15, no. 9, pp. 1179 – 1192, 2007.

[13] L. Lasdon, S. Mitter, and A. Waren, "The conjugate gradient method for optimal control problems," *IEEE Trans. on Automatic Control*, vol. 12, no. 2, pp. 132–138, Apr 1967.

[14] L. Lasdon, "Conjugate direction methods for optimal control," *IEEE Trans. on Automatic Control*, vol. 15, no. 2, pp. 267–268, April 1970.

[15] V. H. Quintana and E. J. Davison, "Clipping-off gradient algorithms to compute optimal controls with constrained magnitude," *Int. J. of Control*, vol. 20, no. 2, pp. 243–255, 1974.

[16] A. E. Bryson and Y. C. Ho, *Applied Optimal Control: Optimization, Estimation and Control*. CRC Press, USA, 1975.

[17] D. E. Kirk, *Optimal Control Theory: An Introduction*. Dover publication, USA, 2004.

[18] J. T. Betts, *Practical Methods for Optimal Control and Estimation Using Nonlinear Programming*. Cambridge University Press, USA, 2009.

[19] J. A. Snyman, C. Frangos, and Y. Yavin, "Penalty function solutions to optimal control problems with general constraints via a dynamic optimisation method," *Computers & Mathematics with Applications*, vol. 23, no. 11, pp. 47 – 55, 1992.

[20] V. M. Becerra, "Solving optimal control problems with state constraints using nonlinear programming and simulation tools," *IEEE Trans. on Education*, vol. 47, no. 3, pp. 377–384, Aug 2004.

[21] S. Lenhart and J. T. Workman, *Optimal Control Applied to Biological Models*. Boca Raton, Florida: Chapman and Hall/CRC, 2007.

[22] W. Sun and Y. Yuan, *Optimization Theory and Methods: Nonlinear Programming*. Springer, USA, 2006.